SOME THOUGHTS ON 44 YEARS OF INTERACTIVE MUSIC MAKING

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ABSTRACT

In 1968, as an undergraduate student at Albany, my teacher Joel Chadabe introduced us to a new concept he'd been working on: interactive composing. Not without some questioning (how was this different from collective improvising? Or mowing the lawn?), I accepted this concept, and have been making interactive works in one form or another ever since. Today, I see the concept of interactivity as having pervaded almost every aspect of our society. In this paper, I will look at a number of pieces I've composed using this idea.

1. INTRODUCTION

Back in 1968, as an undergraduate student at Albany, my teacher Joel Chadabe introduced us to a new concept he'd been working on: interactive composing. Not without some questioning (how was this different from collective improvising? Or mowing the lawn?), I accepted this concept, and have been making interactive works in one form or another ever since. Today, I see the concept of interactivity as having pervaded almost every aspect of our society. Every time you go to an ATM, or your bank's website, you're experiencing industry's degraded concept of interactivity. And as technology has become faster and more powerful, even things which we held up as not being examples of interactivity back in the 60s – such as simple instrumental multi-track recording, or working with a text-based language such as CSound (or Music II, III or IV back then), or the Composers' Desktop Project–today have substantial elements of real-time interactivity about them. Any current DAW can be used either interactively, as well as linearly, and I'm sure someone will emerge who will be doing real-time live coding with CSound, if they aren't already doing a presentation at this conference. And aside from worrying about definitions, it seems to me that "Interactivity" is to computer music today what "I IV V I" is to the blues: the very structural basis of the form. In fact, if I wanted to be a smart-aleck (moi?) I'd quip that having "interactivity" as a theme of a computer music conference is like having "the sunrise" as a theme for a meteorology conference – interactivity is just that pervasive, as far as I can see, in computer music today.

But there are many different kinds of interactivity. And over the years, I've done many collaborative performances which used interactive processes in unique ways. I want to talk about several of these, describing how they worked and what the interactive elements were.

2. WEB (1972)

In chronological order, let's begin with *WEB (1972)*, which was a collaborative performance with Buchla synthesizers and tape delays, performed on Nov. 19, 1972, by Warren Burt, Ernie Morgan, and Bruce Rittenbach, at the University of California, San Diego, under the patient mentorship of Pauline Oliveros. This was a 4 channel performance with 3 performers. Each performer had a Buchla Box in front of them, which included a sequencer, and which fed into their own tape delay system, and in which sounds and control signals from one synthesizer was fed to the next synth in sequence, 1 to 2, 2 to 3, and 3 to 1. Each player had the option to work or not with the control and sound signals from the adjacent synth. So it was almost never a feedback loop, but did allow for some electronic interaction between the players. Each player also was playing with a patch in which the sequencer was semi-unpredictably controlling their sounds. So playing the final result was a bit like surfing an only quasi-predictable structure. Actually, though, despite this being San Diego, California, only one of us – Ernie Morgan – was actually a surfer, and at a semi-professional level, at that. Despite the potential for chaos, listening back to the performance after 40 years, I'm surprised at how gentle and sparse the performance is.

SOUND EXAMPLE – "WEB" (1972), by Warren Burt, Ernie Morgan, and Bruce Rittenbach

3. SENSUS (1988)

In 1987, I began working with Simon Veitch, inventor, with his 3DIS system. This was a system that used video cameras to detect motion and which then produced MIDI signals to control sound. The first large scale use of the system was an installation called *SENSUS (1988)* which took place at Expo 88, in fact, just a couple of hundred metres from this very spot, in the Southbank Parklands. Our pavilion was very near the Nepalese Pagoda, and in fact, when William Duckworth staged his "iOrpheus" here in 2007, he had the wedding of Orpheus and Euridice take place outside the Nepalese Pagoda, and he asked me to do a sound background for the wedding, to

be made of the sounds of the Sensus installation. So the sounds that had filled the air here in 1988 returned in 2007, to be part of an outdoor opera on the same site.

For the Sensus installation, there were about 40 places where the public could interact with it. Each interaction was very simple – someone moved near a camera, in a specific space, and a particular sound was triggered off. Some of these were inadvertent – walking through the entrance arches triggered off a choice of bird samples, and others were triggered off only by wilful gestures – such as the sound of breaking glass triggered off by sticking your hand through an empty window pane. A few sounds even allowed some very simplistic performing. Three hollow cubes, with metal frames, had pentatonic scales of samples of vibes, bass and drums, so three people could improvise a kind of pentatonic jazz, if they wanted to. As said, each interaction was very simple. What made the installation fun (and even qualify as new music), was the fact that there were so many of them, and the sound vocabulary was extremely varied. With even 3 or 4 people in the installation, things got thick very fast. And when there was a crowd around (which there usually was), the sound world was about as complex as Cage's *Williams Mix.* What kept me interested was that usually each participant was fixated on their own sound interaction, and wasn't giving attention to the sound installation as a whole, meaning the overall sound was a chance-produced result of a number of people playing at the same time.



Figure 1. Details of "SENSUS" installation, Expo88, Brisbane, by Warren Burt and Simon Veitch

SOUND EXAMPLE: "Sensus" 1988

4. HEAR THE DANCE, SEE THE MUSIC

The next year, 1989, we used 3DIS in a work for dancers. Funded by Arts Victoria and Diabetes Australia, *Hear the Dance, See the Music: Fair Exchanges (1989)* (Burt, 1989) was a collaboration between Simon Veitch (technologist), myself and Ros Bandt (composers) and Sylvia Staehli, Jane Refshauge and Shona Innes (choreographers and dancers). The program consisted of five pieces, each of which used the system in different ways. Some of the pieces were very simple in their interaction – a moving body triggered off instrumental sounds at particular places, and the dancer could play music in very traditional ways, such as Jane Refshauge's *Inside Out (for Graham)*. Other pieces such as *Free Trade Zones,* by Sylvia Staehli and me, involved elements of unpredictability in the interaction. In this piece, a political comment on the use of exploited female labour in the third world by the electronics industry, Silvia walked around an invisible figure 8 pattern on the floor. At certain points she would come to a sensitive spot. This sensitive spot had a "percentage chance" (say 40%) of triggering on a sample of "ethnic music." If she did trigger off a sound, she danced to it, and each time she passed through the area, she also had a certain chance of turning off the sound. A particular music would only happen in a particular place, but the turning on and off of the sound was probability based, and not by means of a simple on-off switch.



Figure 2. Sylvia Staehli performs "Free Trade Zones"

VIDEO EXAMPLE: "Free Trade Zones," by Burt and Staehli

In another piece, *For Percy Grainger and Burnett Cross* by myself and the three dancers, we attempted to make a dancer-controlled version of Grainger's Reed Box Tone Tool - a microtonal harmonium that used paper rolls to produce glides of closely tuned harmonium reeds. For our piece, we made a very large invisible S-shaped pattern on the floor. Low pitches were at one end, and high pitches at the other. If one walked along the length of the S, one could produce a 2 octave glide of samples of the original reedbox tuned to 36 notes per octave. If one walked across the S, one would trigger off, say, some low notes, then some middle-range notes, then some high notes. If you freely moved around the space, one could produce all manner of musical phrases, most of which (because of the layout of the floor) would have some aspect of gliding about them. This video excerpt begins with all 3 dancer on chairs in the middle of the S. After a bit of a "mad scene" in which Jane and Shona move around the chairs, then move out of the range of the cameras, Sylvia proceeds to roll back and forth along the row of chairs, producing mid-range glides. Eventually, all three dancers end up waltzing around the

space holding chairs, producing a wide ranging texture of gliding and stable notes.



Figure 3. Silvia Staehli, Jane Refshauge, Shona Innes in "For Percy Grainger and Burnett Cross"

VIDEO EXAMPLE: “For Percy Grainger and Burnett Cross” by Burt, Staehli, Refshauge and Innes.

5. HORIZON RERUN

One of my longest running artistic collaborations has been with choreographer Eva Karczag. We have been working together, off and on, since 1977. In 1994, we produced an hour-long dance and music piece called *Horizon Rerun*, which we performed in Melbourne, Sydney and Brisbane. For one section of the dance, an interesting interactive process was used. I had a Buchla Lightning – a device which used the motion of infrared sensors to produce MIDI signals to control electronic music devices. In my case, the infrared sensors were worn on my hands in the form of rings. My movements controlled a microtonal sampled electric piano. The process was this: Eva would be moving. With my hands, I would try to follow her movement. This would produce the microtonal electric piano music. She had the option of responding to the music, or not. If she did, the movement of my hands would produce different music, which she was then free to respond to or note, etc. So we had the potential for a feedback loop, mitigated by her decision whether to respond to the music or not, and my ability to follow her movements accurately with my hands. Unfortunately, no video record of this performance exists, but we do have photos from the Brisbane performance, and a sound recording.



Figure 4. Eva Karczag and Warren Burt in "Horizon Rerun" Queensland Art Gallery 1994



Figure 5. Eva Karczag in "Horizon Rerun" Queensland Art Gallery 1994

SOUND EXAMPLE: “Horizon Rerun” by Burt and Karczag

6. INSTALLATION FOR 3 LAPTOPS: REMEMBERING KENNETH’S LEMONS

The *Installation for 3 Laptops (Remembering Kenneth’s Lemons)*, from 1999, made for the “First Iterate” conference at Monash University, used a different type of interactivity. Three laptops, 2 DOS machines and one early Windows XP machine were connected in series – A to B, B to C and C to A. All machines were running a kind of chaotic generative process based on counters and shift registers, and they were all connected in a feedback loop. Two of the machines were running John Dunn's Music Box, one of the earliest algorithmic “tool kit” programs, and one was running his SoftStep, a further development of the program within a GUI environment. All three computers controlled microtonal electric piano patches. The title is a reference to Kenneth Gaburo’s early 1960s electronic composition, *Lemon Drops* which also used electronic means (Harmonic Tone Generator and splicing) to assemble an imaginary electronic piano

soundscape. The laptops were put on top of plinths, and mounted in a cross between an art-sculpture and a department store display. A panoply of small stuffed animals assisted the installation as well.



Figure 6. Installation for Three Laptops (Remembering Kenneth's Lemons) First Iteration conference, 1999

SOUND EXAMPLE: "Installation for 3 Laptops (Remembering Kenneth's Lemons)" (1999)

7. ROCK AND LIGHT TRACING, BY CATHERINE SCHIEVE

Catherine Schieve's *Rock and Light Tracing (2006)* was a piece for the reconstructed Grainger-Cross Electric Eye Tone Tool, which I constructed in 2004, and rebuilt in 2006. In this piece, the transparent plastic sheeting which constituted the score for the machine was made by placing the plastic directly onto some shale rocks in Newfoundland, (where we were performing the piece) and painting the patterns of the rocks on them. In performance, the score is not only moved along the photocells, in the manner of a traditional player piano score, but also moved in the air above and around the machine, creating highly unpredictable, but very attractive patterns of sound. By moving the score into the 3rd dimension, above the photocells, we were able to easily get beyond Grainger's 2 dimensional conception of the score, as well as having a lot of fun.



Figure 7 .Catherine Schieve: "Rock and Light Tracing" detail of score 2006



Figure 8. Warren Burt and Catherine Schieve perform "Rock and Light Tracing" Sydney 2008

VIDEO EXAMPLE: "Rock and Light Tracing" by Catherine Schieve, Warren Burt assisting.

8. ROBOJAM 2

In 2010, I participated in the second Logos international jam over the internet controlling the Logos Robot Orchestra (Logos 2012). The Robot Orchestra was built by Godfried Willem Raes at the Logos Foundation in Ghent Belgium. It consists of over 40 acoustic instruments, rigged up with solenoid control of various kinds, controlled by MIDI. For the Robo-Jams, Kristof Lauwers of Logos arranged a chat-room style interface, where international participants could type in strings of characters. These gave the central computer, at Logos, materials with which it could control the performers' selections of the Logos instruments. So each performer around the world could specify an instrument, and algorithmic instructions indicating how the instrument was to be played. In the chat-room, we could all see what was happening, and we could all hear the composite piece, which was also occurring live in a concert at Logos. In Robo-Jam 1, in 2008, internet speeds were very slow, and the improvising was difficult. By 2010, internet speeds were higher, and we could use Skype to hear the overall sound, so it was much easier. Also, none of us were specifying musical instructions that required precise rhythmic sync with the other players, so the small delays we experienced were not any significant barrier to doing a good performance. The players for this one were Kristof Lauwers, Yvan Vandersanden and Troy Rodgers at Logos Foundation, Gent, Belgium; Celio Vasconcelos at home in Aachen Germany; Scott Barton and Steve Kemper at the University of Virginia (Virginia Center for Computer Music) Charlottesville, Virginia USA; Jaime Reis in Linda-a-Velha, Portugal; Warren Burt at home in Wollongong Australia; Simon Halsberghe - in a lazy armchair in Antwerp Belgium; Brent Wetters - at home in Providence, Rhode Island, USA; and Juan Sebastian Lach Lau - at home in Morelia, Michoacan, Mexico.

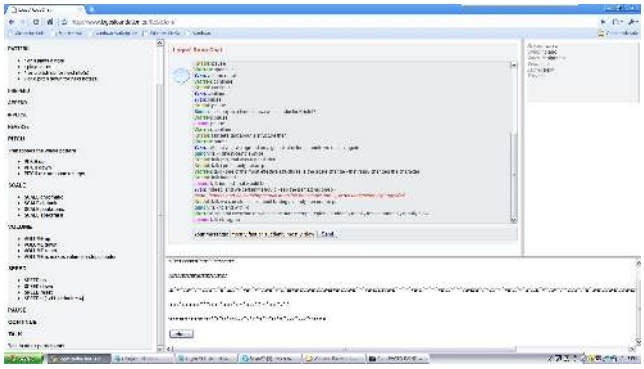


Figure 9. Example of the Chatroom like interface developed by Kristof Lauwers for RoboJam 2



Figure 10. Jaime Reis performing in RoboJam2 in Lisbon



Figure 11. QT a quarter tone organ, one of the Logos Musical Robots



Figure 12. Vibi, a computer controlled vibraphone, one of the Logos Musical Robots

SOUND EXAMPLE: “Robojam 2”

9. THREE MOVEMENTS FOR FOKKER ORGAN

My most recent international interactive performance was *3 Movements for Fokker Organ (2011)* which took place between my home in Daylesford, and the Fokker Microtonal Organ of the Huygens-Fokker Foundation in Amsterdam. The Fokker organ was designed and built by Adriaan Fokker in the early 1950s (Stichting Huygens-Fokker 2012). It is a 31-tone per octave organ, with 2 manuals and pedals. About 5 years ago it was removed from the Teyler's Museum in Haarlem, into the Muziekgebouw aan t'Ij, in Amsterdam and MIDI control was added to it. With the assistance of Ere Lievonen, who is the principal organist for the Fokker Organ and Danny De Graan and Tim Opie, who developed various versions of the Max Patch which enabled us to send midi over the internet, I played 4 algorithmic programs, written with John Dunn's ArtWonk, which generated MIDI which was then sent over to the Fokker Organ. I heard the results back via Skype. The total delay was about 1 second, which was easily good enough to control an interactive algorithmic process. My favorite of the pieces was a little encore movement, *Fokker Organ Wren*, which used DNA protein patterns from the Brazilian Uirapuru, or Organ Wren, as a basic sequencer material, which I elaborated upon in real time.



Figure 13. Warren Burt (on video screen) and Ere Livonen (at the Fokker Organ) performing Burt's "Three Movements for Fokker Organ" Muziekgebouw Amsterdam 2011



Figure 14. Warren Burt performing "Three Movements for Fokker Organ" Daylesford, Vic. 2011 - netbook is controlling the organ, laptop is used as Skype interface to hear organ and watch concert hall.

SOUND EXAMPLE: Excerpt from "Fokker Organ Wren" (2011)

10. CONCLUSION

In each of these pieces then, there is a different kind of interactivity happening. Some are algorithmic, some involve people processes, and some involve remote interaction. The variety of processes used in these pieces shows the wide range of activities that the notion of interactivity can embrace. As they used to say, it truly is a very broad church. But all of them, whether involving people or machines, involve what is an essential element of interactivity – the use of unpredictability as an essential element in musical performance.

11. REFERENCES

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